Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with 2+ years building production cloud infrastructure, distributed deployment automation, backend services, and Kubernetes platforms using Java, Rust, Python, Go, AWS, and CI/CD. Current research in AI powered Android test automation.

EDUCATION

Master of Science in Computer Science Aug 2024 to Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Database Theory, Intelligent Data Analysis.

Bachelor of Engineering in Computer Engineering Aug 2018 to Jul 2022
AISSMS College of Engineering
Relevant Coursework: Object Oriented Programming, Data Mining, Machine Learning.

TECHNICAL SKILLS

Programming: Java, Rust, Python, Go, TypeScript, JavaScript, C/C++, SQL.

Web & Backend: REST APIs, Microservices, Distributed Systems, API Design, Node.js, FastAPI, Flask, React.

Databases: PostgreSQL, MySQL, Elasticsearch, Redis, MongoDB, Vector Databases.

Developer Tools: Git, GitLab CI/CD, Jenkins, AWS (EC2, S3, Lambda, SQS), Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash.

AI & Data: PySpark, Delta Lake, Gemini, Vertex AI, PyTorch, scikit-learn, LangChain, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 to Present

- Designed memory augmented LLM pipeline for Android bug reproduction, separating video understanding from automation across **53 runs**.
- Reduced average run latency **25.7x** (39.4s vs 1012.1s) while preserving task completion fidelity across **20+ apps**.

Software Development Engineer | Siemens Industry Software Jul 2022 to Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python and Boto3, cutting deployment time **40%** across AWS environments.
- Executed **44 production upgrades** in 4 months, debugging Linux, EC2, database, and service issues in live environments.
- Designed Ansible Blue Green deployment framework with rollback support, reducing production upgrade downtime by **50%**.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and Vault.
- Increased automated code coverage from **35% to 78%** as Quality Coach; mentored **5 engineers** on standards and reviews.

Software Developer Intern | Siemens Industry Software Mar 2022 to Jun 2022

- Automated AWS provisioning with Terraform and Go Terratest, cutting deployment time **30%** and catching **95%** misconfigurations.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 to Dec 2021

- Built React and MySQL workflow app digitizing record management and improving departmental operating efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, Delta Lake, FastAPI, PostgreSQL, Redis] Jul 2026

- Built Spark data pipeline forecasting demand across **263 zones** from **2.3M** hourly rows; served predictions through FastAPI.
- Improved MAPE from **18.77% to 16.52%** and cut RMSE **43%** with a two stage Poisson hurdle model.

PolicyPilot Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Built TypeScript REST API and **17 tool MCP server** with Zod validation for structured policy search and comparison.
- Engineered citation backed RAG pipeline with version tracing, reducing hallucination risk in high stakes eligibility answers.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).